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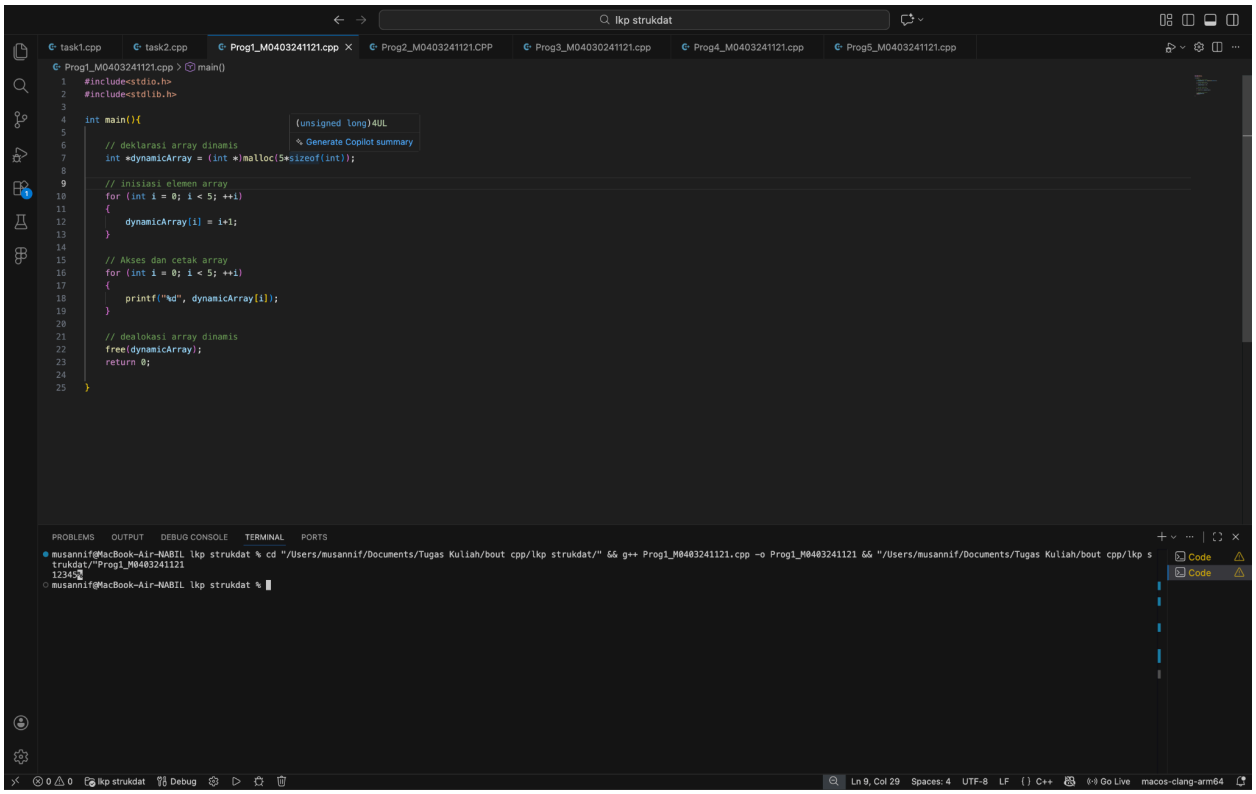
NIM : M0403241121

LKP 2

1. Prog1_M0403241121.cpp



2. Prog2_M0403241121.cpp

```c
#include<stdio.h>
#include<stdlib.h>

int main(){

    // deklarasi array dinamis
    int *dynamicArray = (int *)malloc(5*sizeof(int));

    // inisiasi elemen array
    for (int i = 0; i < 5; ++i)
    {
        dynamicArray[i] = i+1;
    }

    // Akses dan cetak array
    for (int i = 0; i < 5; ++i)
    {
        printf("%d", dynamicArray[i]);
    }
    printf("\n");
    // penambahan alokasi array di tengah program
    int newSize = 8;
    dynamicArray = (int *) realloc(dynamicArray, newSize * sizeof(int));

    // inisiasi elemen tambahan
    for (int i = 5; i < newSize; ++i)
    {
        dynamicArray[i] = i+1;
    }

    // Akses dan cetak array tambahan
    for (int i = 5; i < newSize; ++i)
    {
        printf("%d ", dynamicArray[i]);
    }

    // dealokasi array dinamis
    free(dynamicArray);
    return 0;

}
```

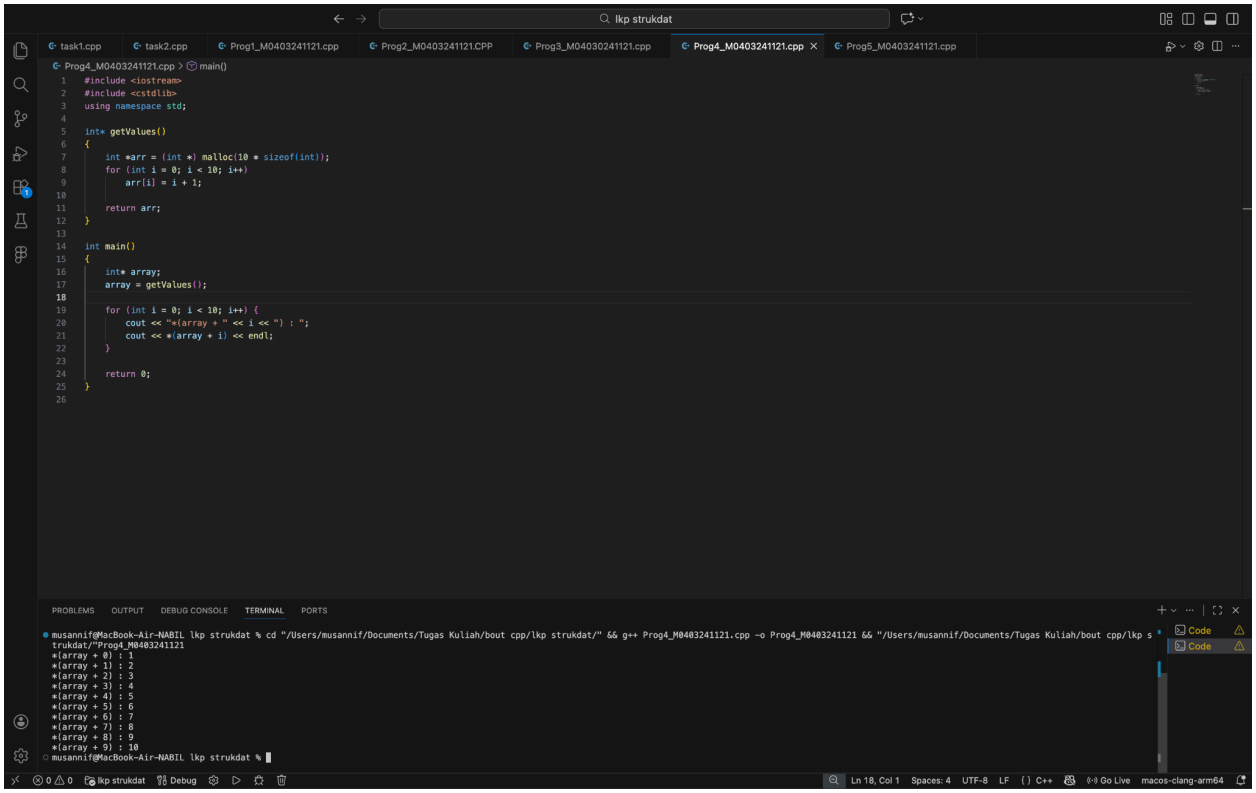
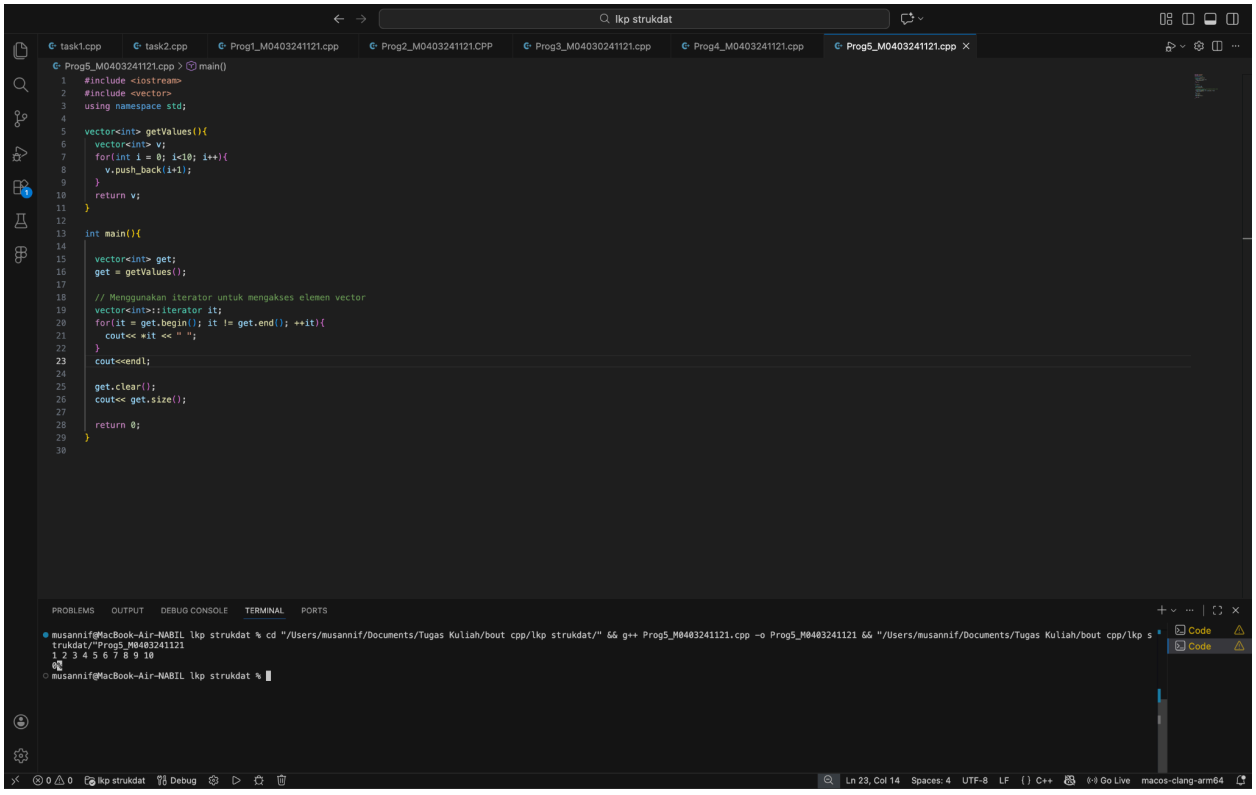
```
musannif@MacBook-Air-NABIL lkp strukdat % cd "/Users/musannif/Documents/Tugas Kuliah/bout cpp/lkp strukdat/" && g++ Prog2_M0403241121.CPP -o Prog2_M0403241121 && "/Users/musannif/Documents/Tugas Kuliah/bout cpp/lkp s
trukdat/"Prog2_M0403241121
12345
6 7 8
musannif@MacBook-Air-NABIL lkp strukdat %
```

3. Prog3_M0403241121.cpp

```c
#include<stdio.h>
#include<stdlib.h>

int main(){

    int n,i;
    int *a, *b;
    scanf("%d", &n);

    a = (int *) malloc (n*sizeof(int));
    b = (int *) malloc (n*sizeof(int));

    for (int i = 0; i < n; i++)
    {
        *(a+i) = rand()%10;
        *(b+n-i-1) = *(a+i);
    }

    for (int i = 0; i <n; i++)
    {
        printf("%d ", *(a+i));
    }
    printf("\n");
    for (int i = 0; i <n; i++)
    {
        printf("%d ", *(b+i));
    }

    free(a);
    free(b);
    return 0;

}
```

```
musannif@MacBook-Air-NABIL lkp strukdat % cd "/Users/musannif/Documents/Tugas Kuliah/bout cpp/lkp strukdat/" && g++ Prog3_M04030241121.cpp -o Prog3_M04030241121 && "/Users/musannif/Documents/Tugas Kuliah/bout cpp/lkp
 strukdat/"Prog3_M04030241121
5
7 9 3 8 0
0 8 3 9 7
musannif@MacBook-Air-NABIL lkp strukdat %
```

4. Prog4_M0403241121.cpp



5. Prog5_M0403241121.cpp